# Programming Fundamentals II: Lab 06

**Due Date: Tuesday, October 22 at Midnight**

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## Part 1. Array Index Exceptions (10 Marks)

As we know, in Java if you try to access an invalid index of an array, an exception will occur. We have two cases here; One where the index is greater than (or equal to) the length of the array, and another where the index is below zero.

a. **(3 Marks)** Create two new exception classes to cover these cases called ArrayIndexUnderflowException and ArrayIndexOverflowException. These exceptions should be specialized versions of ArrayIndexOutOfBoundsException. Have their constructors accept two arguments and store them in private variables: the index that was being accessed, and the length of the array.

b. **(3 Marks)** Create a class called Driver. Add a method called accessElement. This element will accept two arguments: an array of ints, and an int index. This method will try to access the specified element in the array. If the index is too high, throw an ArrayIndexOverflowException, if it's below zero, throw an ArrayIndexUnderflowException. If the index is valid, just return the element at that index.

c. **(4 Marks)** Add a main method to the driver class that will test our exceptions. Declare an array of ints that holds the numbers from 0 to 50 (inclusive). In a loop, generate 100 random ints between -100 to 100, and try to access the array at that index. Catch any exceptions that occur and print "Underflow" if there's an ArrayIndexUnderflowException, or print the last element of the array if there's an ArrayIndexOverflowException. If there's no exception, just print the returned value.

**Sample run**

```
testArray[-59]: Underflow!
testArray[38]: 38
testArray[-49]: Underflow!
testArray[4]: 4
testArray[-33]: Underflow!
testArray[-35]: Underflow!
testArray[-98]: Underflow!
testArray[-88]: Underflow!
testArray[-55]: Underflow!
testArray[-75]: Underflow!
testArray[99]: 50
testArray[46]: 46
testArray[-37]: Underflow!
testArray[-19]: Underflow!
testArray[99]: 50
testArray[30]: 30
testArray[44]: 44
testArray[-64]: Underflow!
testArray[7]: 7
testArray[23]: 23
testArray[94]: 50
testArray[-44]: Underflow!
…
```

Add the **date and your name and student number** to the top of your java files. Please submit your java files to Nexus before the end of the day the lab date. **Late submissions will be not be accepted.**